



Optimizing PMSM performance by integrating predictive current, speed and sliding mode flux weakening controllers

Otimizando o desempenho do PMSM integrando controladores preditivos de corrente, velocidade e enfraquecimento de fluxo em modo deslizante

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ABSTRACT

To improve the control performances of permanent magnet synchronous motors (PMSMs), this paper combines the sliding mode controller (SMC) and model predictive controller (MPC) to balance the contradictions between stability and dynamic performance, reduce current and torque ripples, and provide precise, complete current and speed control. At first, the MPC is employed to control the current and speed of motors, which could solve the chattering issue of the traditional sliding mode control method and effectively guarantee the stability of the control system. Second, a sliding mode observer (SMO) is incorporated to eliminate the interference of flux variation and then reduce the current and torque ripples. Meanwhile, the sliding mode controller is used to adjust the speed of motors and generate an adjustable switching surface to implement rapid and accurate control of the motor. This paper presents the design and realization process of each controller in detail, respectively shows the design principle of the combined controller and verifies the control effect of the combined controller under different motor speeds through simulation, and analyzes the control advantages and its application. The simulation results prove that the combined controller reduces the current and torque ripples to a certain extent in speed regulation, and the control performance is more stable and robust over a wide range of speed. In addition, this paper discusses the practical application of combined controllers in industrial and electric vehicle motor control in the later stages and presents the



influence and significance of combined controllers on the efficiency and stability of PMSM control. From the perspective of application and system integration, the integrated design of SMC, SMO, and MPC can further promote the development of high-performance PMSM in the industrial and automotive industries in the future.

Keywords: flux weakening, model predictive control, PMSM, sliding mode observer, sliding mode control.

RESUMO

Para melhorar o desempenho de controle dos motores síncronos de ímã permanente (PMSMs), este artigo combina o controlador de modo deslizante (SMC) e o controlador preditivo do modelo (MPC) para equilibrar as contradições entre estabilidade e desempenho dinâmico, reduzir ondulações de corrente e torque, e proporcionar controle preciso e completo de corrente e velocidade. Primeiramente, o MPC é empregado para controlar a corrente e a velocidade dos motores, o que poderia resolver o problema de trepidação do método tradicional de controle de modo deslizante e garantir efetivamente a estabilidade do sistema de controle. Em segundo lugar, um observador de modo deslizante (SMO) é incorporado para eliminar a interferência da variação de fluxo e então reduzir as ondulações de corrente e torque. Enquanto isso, o controlador de modo deslizante é usado para ajustar a velocidade dos motores e gerar uma superfície de comutação ajustável para implementar um controle rápido e preciso do motor. Este artigo apresenta o processo de design e realização de cada controlador em detalhes, respectivamente mostra o princípio de design do controlador combinado e verifica o efeito de controle do controlador combinado em diferentes velocidades de motor por meio de simulação, e analisa as vantagens de controle e sua aplicação. Os resultados da simulação comprovam que o controlador combinado reduz as ondulações de corrente e torque até certo ponto na regulação de velocidade, e o desempenho de controle é mais estável e robusto em uma ampla faixa de velocidade. Além disso, este artigo discute a aplicação prática de controladores combinados no controle de motores industriais e veículos elétricos em estágios posteriores e apresenta a influência e significado dos controladores combinados na eficiência e estabilidade do controle de PMSM. Do ponto de vista da aplicação e integração de sistemas, o design integrado de SMC, SMO e MPC pode promover ainda mais o desenvolvimento de PMSM de alto desempenho nas indústrias industrial e automotiva no futuro.

Palavras-chave: enfraquecimento do fluxo, controle preditivo de modelo, PMSM, observador de modo deslizante, controle de modo deslizante.

1 INTRODUCTION

Permanent Magnet Synchronous Motors (PMSMs) have gained significant popularity in various applications, such as robotics, automotive, aerospace, industrial, and medical fields, owing to their high power density and superior torque-to-inertia ratio (Benkaihoul *et al.*, 2024; Kethiri; Charrouf, 2023b). In



comparison to induction motors (IMs), PMSMs exhibit enhanced efficiency and a higher power-to-weight ratio (Garcia *et al.*, 2011). Additionally, existing PMSM control methods exhibit specific problems or limitations, such as Parameter Sensitivity, Nonlinear Behavior, Sensor Requirements, High-Speed Operation, Torque, and current Ripple (Wei *et al.*, 2021).

The study aims to address these last two limitations. The effective control of PMSMs has been achieved through the utilization of two prominent control methods: Sliding Mode Control (SMC) and Model Predictive Control (MPC). SMC proves advantageous in applications requiring precise and rapid control of PMSMs, such as robotics and automobiles. Additionally, SMC is well-suited for managing PMSMs in systems that are susceptible to external disturbances, like wind turbines. On the other hand, Model Predictive Control (MPC) offers precise regulation of PMSMs in applications that demand high control accuracy (BABQI, 2022). Combining SMC and MPC techniques allows for improved management of PMSMs. By merging the robustness of SMC and the accuracy of MPC, it becomes feasible to achieve both rapid and accurate control of PMSMs across a wide range of applications. Moreover, this combined approach presents an opportunity to reduce the complexity and cost of the control system.

The main contribution of this study is found in the proposal of a sliding mode for speed control, accompanied by an observer flux sliding mode and a finite control set model predictive current and speed control (FCS-MPCCS). The proposed model will be implemented in MATLAB Simulink, and simulation results will demonstrate the successful management of speed, as well as the reduction of current and electromagnetic torque ripples. The novelty of the work is not immediately evident as predictive and SMCs are widely used in the field. However, the particularity of the concept is added to the use of an SM for speed control, coupled with an observer flux sliding mode and FCS-MPCCS. In the following sections, the relevant literature is examined in Section 2. Section 3 addresses the modeling of the PMSM and the creation of the MPC controller, after which the suggested SM flux observer is designed and the SM for speed control is described. In Section 4, simulation findings are presented, and a discussion of simulation outcomes is provided. Finally, in Section 5, the findings are summarized, limitations are acknowledged, and future study topics are suggested.



2 LITERATURE REVIEW

A study by (Lee *et al.*, 2015), compares MPC and SMC as control techniques. MPC, being model-based, accounts for system state and input limitations, making it suitable for applications with stringent control objectives. SMC, on the other hand, is a nonlinear control approach that excels in dealing with parameter fluctuations and external disturbances, albeit with fewer state and input limitations. Depending on the application and desired performance, both MPC and SMC can be effective control methods for PMSMs.

A research by (Liu *et al.*, 2016), proposes the combination of MPC and SMC to leverage their respective advantages. MPC provides high-precision control, while SMC mitigates the effects of parameter changes and unmodeled dynamics. The authors emphasize the need for careful design of the combined controller to prevent complexity and potential instability. Proper evaluation and selection of design parameters are crucial to harness the benefits of the combined approach while mitigating its drawbacks.

The authors (Liu *et al.*, 2017), evaluate the benefits and limitations of combining MPC and SMC for PMSM control. The study finds that the combination reduces torque ripple, improves dynamic performance, and lowers harmonic content in the output. However, it also increases the complexity of the control system and the computation time, potentially leading to slower controller response. Careful consideration of the trade-offs is necessary before employing this strategy.

MPC and SMC can be effectively applied to various types of PMSMs, as demonstrated by (Sreejith; Singh, 2018). They assess the performance of both control methods on different types of PMSMs and find that SMC is more efficient for regulating the speed of PMSMs with a single-rotor winding, while MPC performs better with PMSMs having multiple-rotor windings. Choosing the appropriate control strategy based on the specific PMSM type is crucial.

A comprehensive literature study conducted by (Zaihidee; Mekhilef; Mubin, 2019) highlights the pros and cons of MPC and SMC. MPC excels in tracking accuracy and resilience to parameter fluctuations and disturbances, while SMC is computationally efficient and suitable for low-cost applications. The study suggests that combining SMC with feed-forward control enhances tracking accuracy and robustness, while MPC benefits from the addition of a feed-forward controller to



reduce computational complexity. Combining both techniques, such as the hybrid MPC-SMC method, could offer optimal performance for PMSM control.

In a review conducted by (Xu *et al.*, 2019). The performance, robustness, and complexity of MPC and SMC for PMSM control are thoroughly examined. MPC outperforms SMC in overall performance, but SMC exhibits better resilience to external disturbances. The choice of control method depends on the application and the trade-off between performance, robustness, and complexity. This review serves as a valuable resource for engineers seeking guidance on selecting between MPC and SMC for PMSM control in different applications.

Combining both MPC and SMC provides enhanced resilience and higher performance, according to (Spasic, 2019), MPC predicts future states using models and optimization, while SMC corrects disruptions during operation. The combination can minimize computing complexity by selectively utilizing each approach. MPC and SMC have been increasingly employed to improve PMSM performance by enhancing speed and torque accuracy in the presence of disturbances

According to a study by (Wang; Li; Liu, 2019), the utilization of MPC and SMC has been on the rise to enhance the performance of Permanent Magnet Synchronous Motors (PMSMs). These control techniques offer the ability to improve motor speed and torque accuracy, particularly when faced with external disturbances. MPC enables the anticipation and adjustment of motor speed and torque based on current and predicted environmental conditions. In addition, SMC can be employed to adapt motor speed and mitigate the impact of external disruptions. The authors suggest that combining MPC and SMC has the potential to enhance the performance of PMSMs.

The authors (Sreejith; Singh, 2020), investigates the benefits of using MPC and SMC for PMSMs. The study reveals that both techniques offer various advantages, including enhanced dynamic performance, resilience to parameter fluctuations, and reduced control signal chattering. The authors emphasize the complementary nature of MPC and SMC, which can lead to improved speed and robustness when combined.

A comparative study conducted by (Li *et al.*, 2021), assesses the performance of MPC and SMC in PMSM control. Simulations show that MPC



outperforms SMC in terms of precision, settling time, and maximum overshoot. Due to its superior performance, the authors recommend MPC as the preferred control approach for PMSMs.

In (Hou; Ma; Wang, 2022), the study explores the concept of merging MPC and SMC for PMSMs to achieve an efficient and robust control system. The combination enables greater accuracy, shorter reaction times, and resilience to uncertainties and external disturbances. However, implementing this technique requires careful consideration of the system's complexity and associated challenges. Further research is needed to develop a more efficient and robust control system.

A study by (Li *et al.*, 2022), compares the benefits and drawbacks of MPC and SMC for PMSM control. MPC outperforms SMC in altering motor speed and torque while offering better tracking precision. SMC, on the other hand, is easier to deploy and responds faster, showing superior performance in handling parameter uncertainty and external disruptions. The choice between MPC and SMC depends on the specific application and performance requirements.

According to (Wang; Liu, 2023), MPC is ideal for PMSMs due to its tracking accuracy and resilience, while SMC provides robust performance with minimal computation complexity. Depending on cost and performance requirements, either MPC or SMC can be employed as the preferred control approach for PMSMs.

The literature review highlights the benefits and challenges of merging MPC and SMC for PMSM control. The main findings from the reviewed papers suggest that both MPC and SMC can be effective control methods for PMSMs, depending on the specific application and desired performance. MPC has advantages in terms of tracking accuracy, resilience to disturbances, and system stability. It considers system states and input limitations, making it suitable for applications with tough control objectives. However, MPC has higher computational complexity and implementation costs. SMC, on the other hand, offers faster response time, robustness to parameter fluctuations and external disturbances, and lower computational requirements. It is more computationally efficient and suitable for low-cost applications. Nevertheless, SMC may have limitations in applications with a significant number of state and input limitations. Combining MPC and SMC can enhance resilience, higher control precision, and faster reactions. This



combination can strike a balance between the advantages of both methods and improve overall performance. However, there are challenges in designing the control settings and ensuring stability without compromising system complexity.

The review suggests that the choice between MPC and SMC depends on factors such as the application requirements, tradeoffs between performance, robustness, and complexity, as well as the type of PMSM being used. The combination of MPC and SMC, or a hybrid MPC-SMC approach, may offer a higher performance for PMSM control.

3 RESEARCH METHOD

3.1 MODELLING OF THE MACHINE

A surface-mounted PMSM with three-phase stator windings and a sinusoidal flux distribution is considered. The machine is fed by a two-level three-phase inverter as shown in Figure 1, the general electrical and mechanical properties of the PMSMs are incorporated into the state-space model:

$$\frac{di_d}{dt} = -\frac{R_s}{L_d}i_{sd} + \frac{L_q}{L_d}p\omega_r i_q + \frac{v_d}{L_d} \quad (1)$$

$$\frac{di_q}{dt} = -\frac{L_d}{L_q}p\omega_r i_{sd} - \frac{R_s}{L_q}i_{sq} + \frac{v_{sq}}{L_q} - \frac{\varphi_m}{L_q}p\omega_r \quad (2)$$

$$\frac{d\omega_m}{dt} = \frac{1}{J}(1.5 p(\varphi_m i_q + (L_d - L_q)i_d i_q) - T_l \quad (3)$$

Where

L_d , L_q are dq -axis inductance, and $L_d=L_q=L_s$ for a surface-mounted PMSM. ω_r is the rotor mechanical angular speed. T_l is the load torque. J is the overall inertia considering load. R_s is the stator resistance. p and φ_m represent the number of pole pairs and the flux linkage, respectively. The stator voltage equations can be rewritten in dq coordinates:

$$v_d = R_s i_d + L_s \frac{di_d}{dt} - L_s \omega_r i_q \quad (4)$$

$$v_q = R_s i_q + L_s \frac{di_q}{dt} - L_s \omega_r i_d + j\varphi_m \omega_r \quad (5)$$

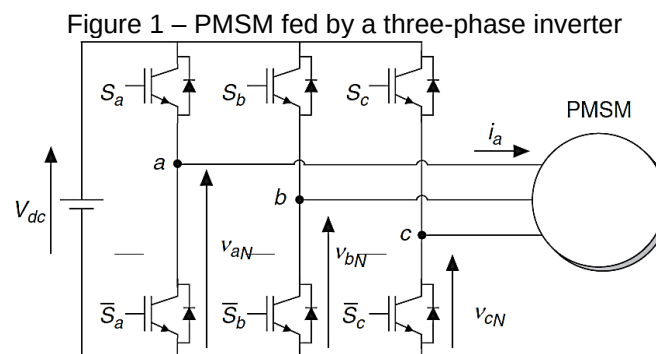
The relation between the stator vectors and the rotating reference frame is shown in Figure 2, the electric torque produced by the machine depends on the flux magnitude and the quadrature component of the stator current vector:

$$T_e = \frac{3}{2} p \phi_m i_q \quad (6)$$

The mechanical rotor dynamics are described by the equation

$$\frac{d\omega_r}{dt} = \frac{1}{J} (T_e - T_l) - \frac{B}{J} \omega_r^m \quad (7)$$

Where B is the friction coefficient, ω_r^m is the rotor shaft's mechanical speed.

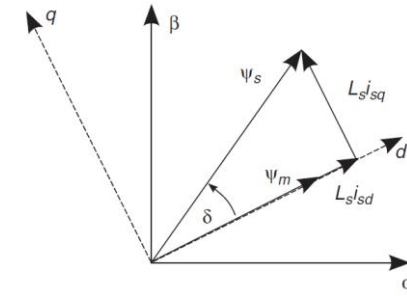


Source: (Rodriguez; Cortes, 2012)

3.2 FIELD-ORIENTED CONTROL USING PREDICTIVE CURRENT AND SPEED CONTROL

Each stator current component is given a physical significance by employing a revolving dq reference frame that is aligned to the rotor magnetic field axis. While the real component i_{sd} is proportional to the reactive power, the imaginary component i_{sq} is proportional to the electrical torque. In this manner, the machine control is carried out as a current control scheme, where the external speed control loop produces the current references. The cost function must take into account the error between the reference currents and anticipated currents while estimating the behaviour of the stator currents using the machine's model. Figure 3, shows a block diagram of the proposed control topology.

Figure 2 – Vector diagram of the stator variables and the rotating reference frame.



Source: (Rodriguez; Cortes, 2012)

3.3 IMPLEMENTATION OF MULTI-OBJECTIVE FCS-MPC

By using the Euler approximation for the stator current derivatives for a sampling time T_s , that is:

$$\frac{di}{dt} \approx \frac{i(k+1) - i(k)}{T_s} \quad (8)$$

The predicted stator currents and rotor speed in the dq reference frame are obtained from Eq. (1), Eq. (2), and Eq. (3) after discretization (T_s) as follows:

$$i_d^p(k+1) = \left(1 - \frac{R_s T_s}{L_s}\right) i_{sd}(k) + \frac{T_s}{L_s} v_d \quad (9)$$

$$i_q^p(k+1) = \left(1 - \frac{R_s T_s}{L_s}\right) i_q(k) - T_s \omega_r i_d(k) - \varphi_m \omega_r T_s + \frac{T_s}{L_s} v_q \quad (10)$$

$$\omega_r(k+1) = \frac{3p\varphi_m T_s}{4J} (i_q(k+1) - i_q(k)) - \frac{T_s}{J} \omega_r(k) + \omega_r(k) \quad (11)$$

These equations enable predictions of the stator currents and speed to be calculated for each one of the seven voltage vectors generated by the inverter.

3.4 HARD CONSTRAINTS

One crucial aspect in the control of PMSMs is the management of hard constraints on internal variables. These constraints ensure that the variables remain within their allowed ranges, preventing any undesirable behavior. In traditional cascaded control schemes, saturation levels are commonly employed to limit the references of these variables. However, in a predictive control scheme,



these limitations can be incorporated differently. To account for hard constraints, an additional term is introduced in the cost function. This term acts as a nonlinear function and generates a significantly high value when the currents exceed the permissible limits. Conversely, it equals zero when the currents are within the allowed range, the function is defined as follows:

$$f_{\lim}(i_s^p) \begin{cases} \infty & \text{if } |i_s^p| > i_{\max} \\ 0 & \text{if } |i_s^p| < i_{\max} \end{cases} \quad (12)$$

Here, i_s^p represents the predicted stator current vector, $i_{\max}$ denotes the maximum allowable stator current magnitude, and $f_{\lim}(i_s^p)$ represents the nonlinear function that incorporates the hard constraints (Rodriguez; Cortes, 2012).

3.5 WEIGHTING FACTOR HANDLING STRATEGY

In the design of the control system, it is essential to consider the balance of state sensitivity to voltage alteration. By analyzing the equations governing the system, it becomes apparent that the currents in the PMSM are more sensitive to voltage changes compared to the speed. To achieve a balanced sensitivity between the two variables, appropriate weighting factors are introduced. These factors ensure that the magnitude of speed and current variations caused by the same voltage change within one control period are at the same level. The weighting factors are defined as follows: From Eq.(10) and Eq.(11), in addition to $uq(k)$, a voltage disturbance $\Delta uq(k)$ is assumed to be applied to the machine at tk . It can be deduced that the current alteration $\Delta iq(k+1)$ caused by $\Delta uq(k)$ in one cycle can be described as:

$$\Delta iq(k+1) = \frac{T_s}{L_s} \Delta uq(k) \quad (13)$$

While the speed alteration $\Delta \omega_m(k+1)$ caused by $\Delta uq(k)$ is:

$$\Delta \omega_m(k+1) = \frac{3p\phi_m T_s}{4J} \Delta iq(k+1) \quad (14)$$



Clearly, the machine's currents are more sensitive to voltage changes. To balance the state sensitivity differences between the two variables, the weighting factors can be defined as:

$$k1 = 1, \quad k2 = \frac{4J}{3p\phi_m T_s} \quad (15)$$

Consequently, the speed and current variations caused by the same voltage change within one control period will stand at the same level. Note that the weighting factor $k2$ is related to the sampling period T_s , representing that the discretization scheme is an important factor that influences the optimization operation of the multi-objective FCS-MPC-based controller.

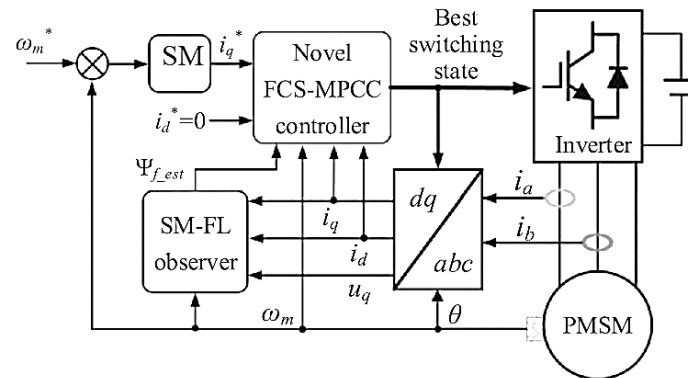
3.6 COST FUNCTION

The evaluation of speed and current performance in the control system is accomplished using a multi-objective cost function. This function assesses the system's performance based on various objectives and is formulated as follows:

$$g = k1 \left(i_{dref} - i_d(k+1) \right)^2 + k1 \left(i_{qref} - i_q(k+1) \right)^2 + k2 \left(\omega_{ref} - \omega_m(k+1) \right)^2 + f_{lim} \quad (16)$$

The cost function incorporates the squared deviations between the reference and actual values of the direct current i_d , quadrature current i_q , and motor speed ω_m . The coefficients $k1$ and $k2$ determine the relative importance of each term in the cost function. Additionally, the f_{lim} term represents the contribution of the nonlinear function that handles the hard constraints. Figure 3, a block diagram illustrates the proposed control topology, showing the components and their interactions.

Figure 3 – Block diagram of the proposed control topology

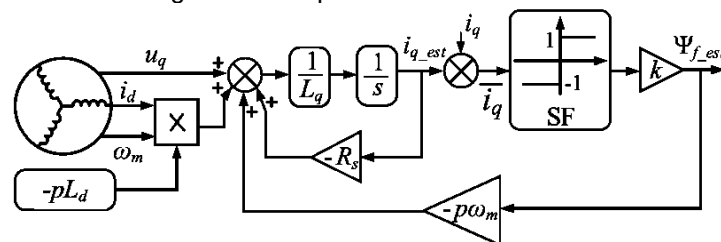


Source: Authors

3.7 PROPOSED SM-FLUX OBSERVER

Sliding Mode control relies on the variable structure control technology, restraining the state variables of a system on the sliding surface (Šabanovic, 2011). SM control has been widely used in parameter identification instances because of its remarkable resilience against external disturbance. In just a few studies does have the flux linkage in PMSM drives been observed using SM theory. According to the machine model, the flux linkage is only included in Eq.(2), according to the theory of SM variable structure, the q-axis current sliding surface S is needed (Han; Gong; Gao, 2022). Figure 4, shows the proposed SM-FL observer.

Figure 4 – Proposed SM-FL observer



Source: (Han; Gong; Gao, 2022)

3.7.1 SM-FL Observer Design

SM-FL observer is specially designed to provide the instantaneous parameter information for the predicting plant. According to the machine model, the flux linkage and according to the theory of SM variable structure, the q axis current sliding surface S_1 is needed:

$$S_1 = \bar{i}_q = i_{q_est} - i_q = 0 \quad (17)$$



Where $i_{q_{est}}$ is the intermediate current state. $\bar{i}_q$ is the error between the estimated current and the real current. the SM observer for flux linkage estimation of the PMSM can be expressed as:

$$\frac{di_{q_{est}}}{dt} = -\frac{L_d}{L_q} p\omega_m i_d - \frac{R_s}{L_q} i_{q_{est}} + \frac{v_q}{L_q} - \frac{p\omega_m}{L_q} kF(\bar{i}_q) \quad (18)$$

Where k is the observer gain. $F(\bar{i}_q)$ is the SF-based switching function, in our case $L_d = L_q = L_s$

$$F(\bar{i}_q) = \text{sign}(\bar{i}_q) \quad (19)$$

When the SM observer gets to the stable state, the real-time flux linkage used for FCS-MPCC equals φ_{f_est} :

$$\varphi_{f_est} = kF(\bar{i}_q) \quad (20)$$

The stability of the SM observer is ensured when the parameter k is set as a positive value greater than the maximum flux linkage. In this study, k is chosen to be twice the flux linkage generated by the permanent magnet (PM) installed in the machine (Yang; Li; He, 2022).

3.7.2 Stability Analysis

In order to analyze the stability condition of the proposed SM-FL observer, a function based on Lyapunov theorem is constructed as follows:

$$V = \frac{1}{2} S_1^T \cdot S_1 = \frac{1}{2} \bar{i}_q^2 > 0 \quad (21)$$

Then, in the light of Lyapunov stability criterion, only when $\frac{dV}{dt} < 0$ will we conclude that the SM observer is stable, take the time derivative of Eq.(21):

$$\frac{dV}{dt} = S_1^T \cdot \frac{dS}{dt} = \bar{i}_q \frac{d\bar{i}_q}{dt} \quad (22)$$

It can be derived according to Eq.(2) and Eq.(18) that:

$$\frac{d\bar{i}_q}{dt} = -\frac{R_s}{L_s} \bar{i}_q - \frac{p\omega_m}{L_s} (kF(\bar{i}_q) - \varphi_m) \quad (23)$$



Substitute Eq.(22) into Eq.(23), it can be further deduced:

$$\frac{dV}{dt} = -\frac{R_s}{L_s} \bar{i}_q^2 - \frac{p\omega_m(kF(\bar{i}_q) - \varphi_m)}{L_s} \bar{i}_q \quad (24)$$

3.8 SLIDING MODE FOR SPEED CONTROL

Sliding Mode Exponential (SME) is an adaptive control method that combines sliding mode control with exponential control. It's known for its optimality, robustness, and reliability in a wide range of applications. Designing SME involves tuning gains for sliding mode and exponential terms to achieve desired performance. The sliding mode term ensures system robustness, while the exponential term enhances transient response and steady-state accuracy (Fallaha; Saad; Kanaan; Al-Haddad, 2010). Choosing an appropriate weighting factor between these terms is crucial for optimal results. SME also requires careful selection of a switching surface to maintain insensitivity to external disturbances, making it an effective and efficient control technique for various applications.

3.8.1 SME Design

The sliding surface S_2 is defined with the motor speed tracking error:

$$S_2 = \omega_{ref} - \omega_m \quad (25)$$

Exponential approach law:

$$\dot{S} = -\varepsilon \operatorname{sign}(S_2) - \gamma S_2 \quad (26)$$

Where the parameters $\varepsilon > 0$, $\gamma > 0$ are a strictly positive real constant, $\operatorname{sign}(S)$ is the sign function

3.8.2 Stability Proof

In order to determine the stability of the controlled system, Lyapunov function was selected:

$$V(x) = \frac{S_2^2}{2} \quad (27)$$



The derivative of it can be obtained:

$$\dot{V}(S) = S_2 \dot{S} = -\alpha_1 S^3 |S_2| \text{sign} - \alpha_2 S_2^2 \quad (28)$$

$(\alpha_1 > 0, \alpha_2 > 0)$

Therefore, $\dot{V}(S_2)$ is less than or equal to 0 for S in any range, which satisfies the stability condition of Lyapunov, and the system is stable.

The Research Method section outlines the methodology for controlling PMSM drives. It begins with the machine modeling and stator voltage equations in dq coordinates. The section proceeds to explain field-oriented control using predictive current and speed control, emphasizing the revolving dq reference frame. It also covers the discrete-time model, equations for predicted stator currents and rotor speed, and the impact of voltage vectors generated by the inverter. Additionally, the section discusses the management of hard constraints and the strategy for handling weighting factors to balance sensitivity between speed and current variations. The cost function for system performance evaluation is described, considering squared deviations and the nonlinear function for hard constraint handling. The section concludes with the proposed SM-Flux Observer, SM-FL observer design, stability analysis, and sliding mode speed control.

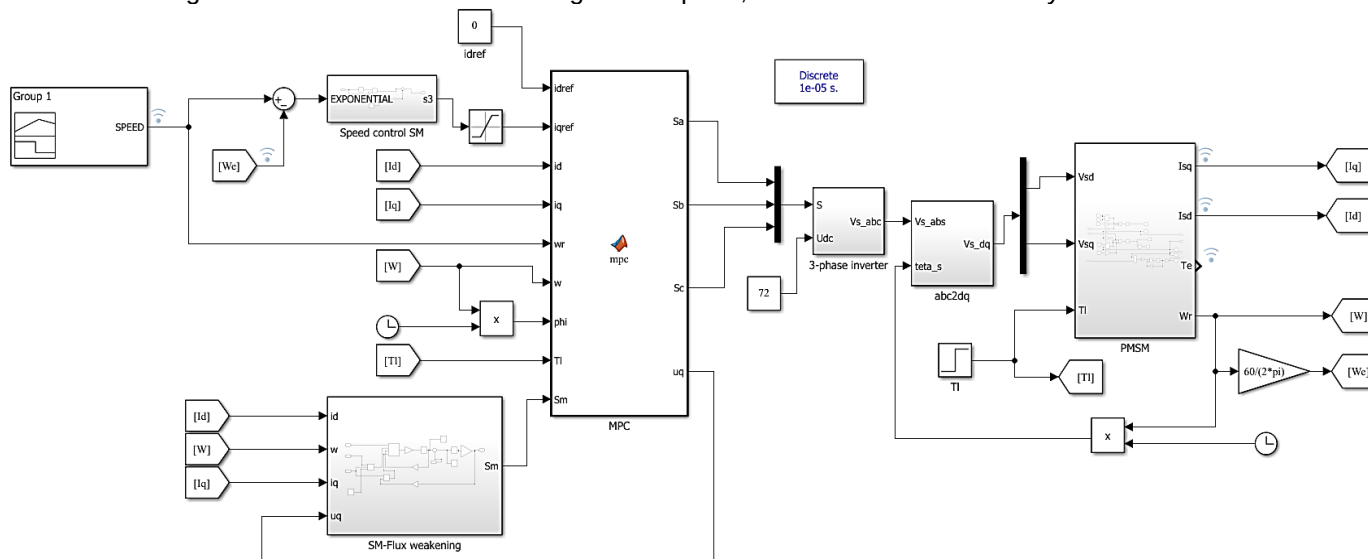
4 RESULTS AND DISCUSSION

In this section a simulations and comparative study is conducted, comparing the proposed control method with another one that does not incorporate the SM-Flux Observer, which can be considered as a traditional method because as it mentioned in section 2, the combination of SMC and MPC are widely used in field of speed control.

In Figure 5, the model and controllers are implemented in MATLAB Simulink. In Figure 6, for different speed regions (divided into three regions), the speed profile is set to 1000 Rpm from 0 to 0.1 s, followed by a decrease to 500 Rpm from 0.1 to 0.2 s, and finally an increase to 1500 Rpm from 0.2 s onwards. Figure 7, provides a comparative analysis of the traditional method and the proposed method in terms of current ripples (QCR). Additionally, Figure 8, presents the electromagnetic torque ripple (TR) for both the traditional and

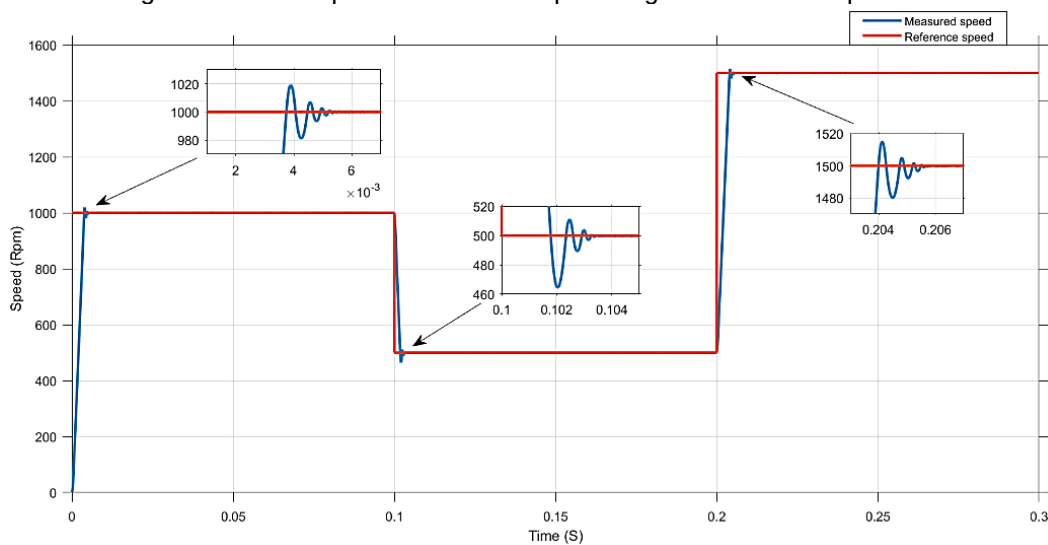
Parameter	Value	Unit
Stator winding resistance R_s	0.485	Ω
dq -axis inductance L_s	0.000395	H
The number of pole pairs p	4	-
Moment of inertia J	0.0027	$\text{Kg} \cdot \text{m}^2$
Damping coefficient B	0.0004924	-
PM flux linkage φ_m	0.1194	Wb
DC-bus voltage V_{dc}	72	V
Maximum current i_{max}	23	A
Sampling time T_s	1e-5	S
Torque load T_l	2	Nm
Observer gain k	0.2388	-
Weighting factor k_1	1	-
Weighting factor k_2	753.76	-

Figure 5 – Simulation structure diagram of speed, current and flux control system for PMSM.



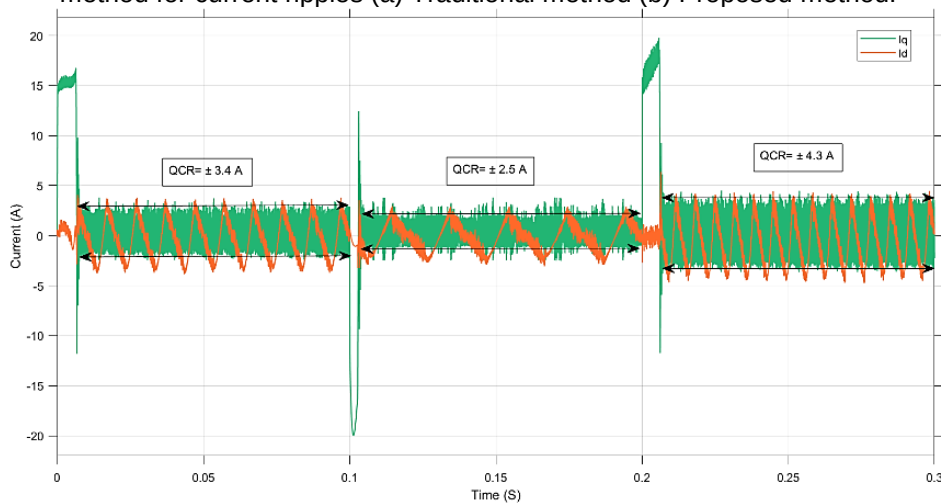
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Figure 6 – Rotor speed for different speed regions without torque load.

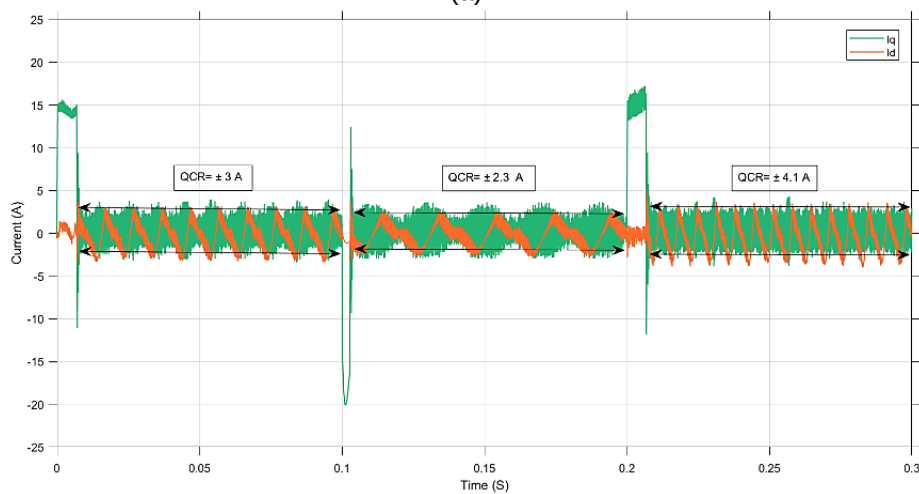


Source: Authors

Figure 7 – Comparison results without torque load of the proposed method and the traditional method for current ripples (a) Traditional method (b) Proposed method.



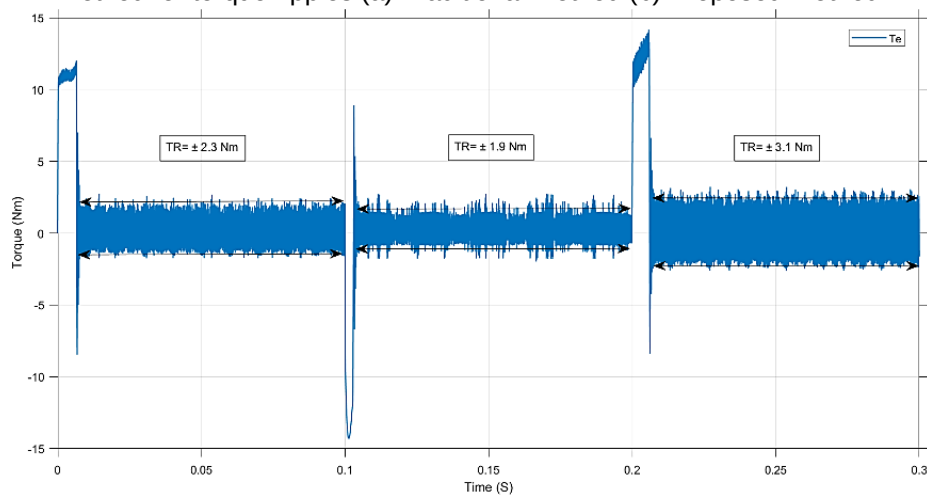
(a)



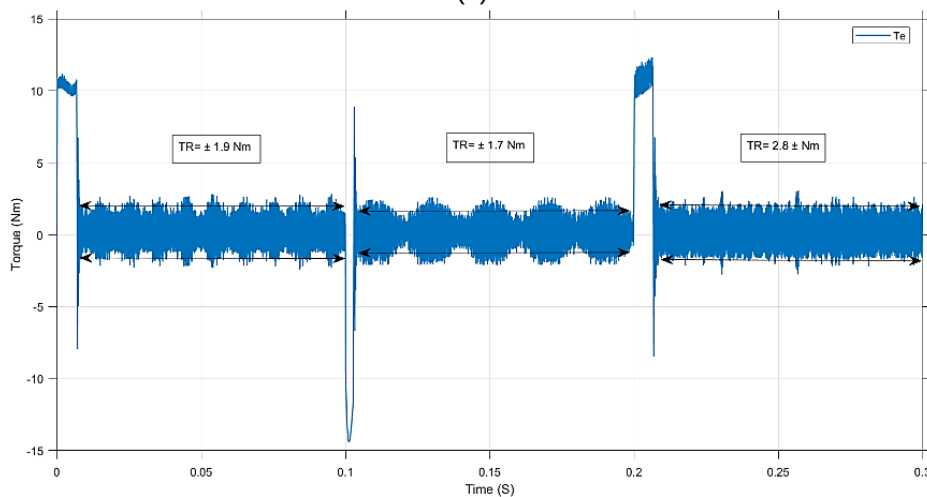
(b)

Source: Authors

Figure 8 – Comparison results without torque load of the proposed method and the traditional method for torque ripples (a) Traditional method (b) Proposed method.



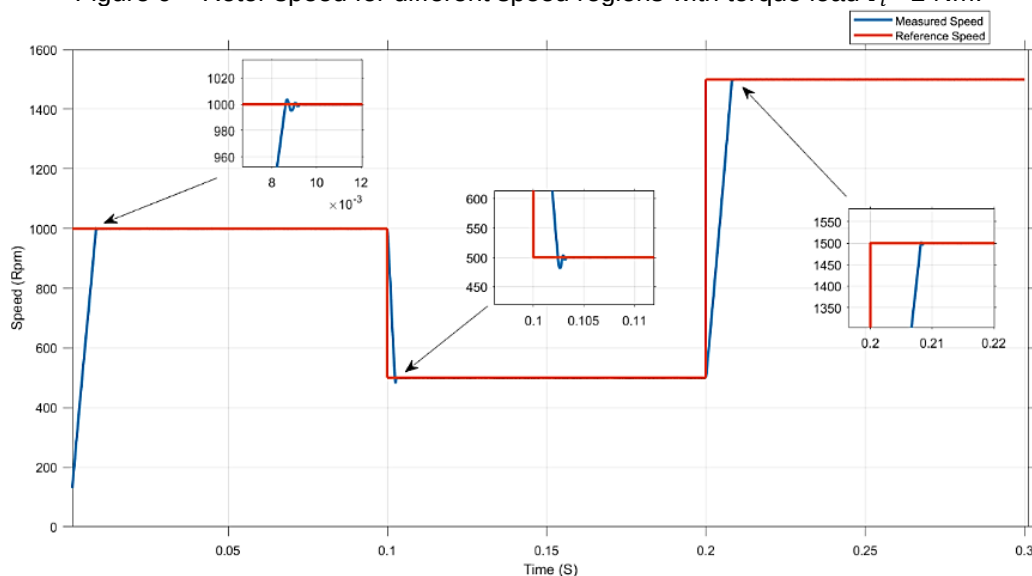
(a)



(b)

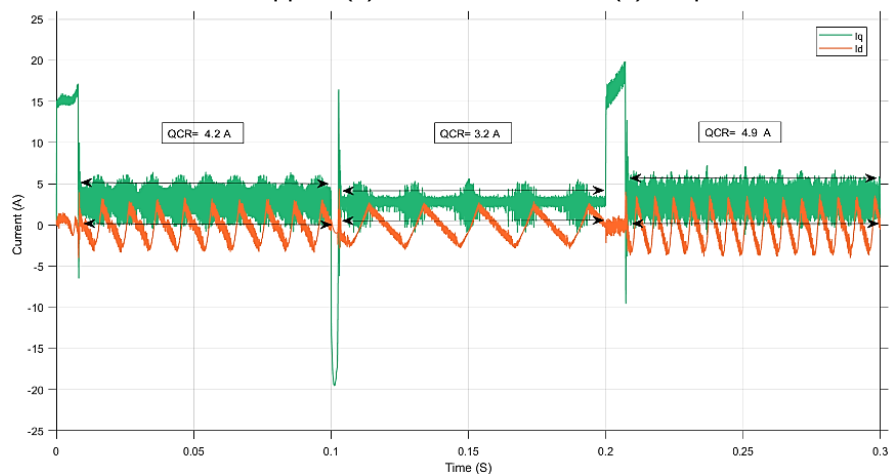
Source: Authors

Figure 9 – Rotor speed for different speed regions with torque load $T_l = 2$ Nm.

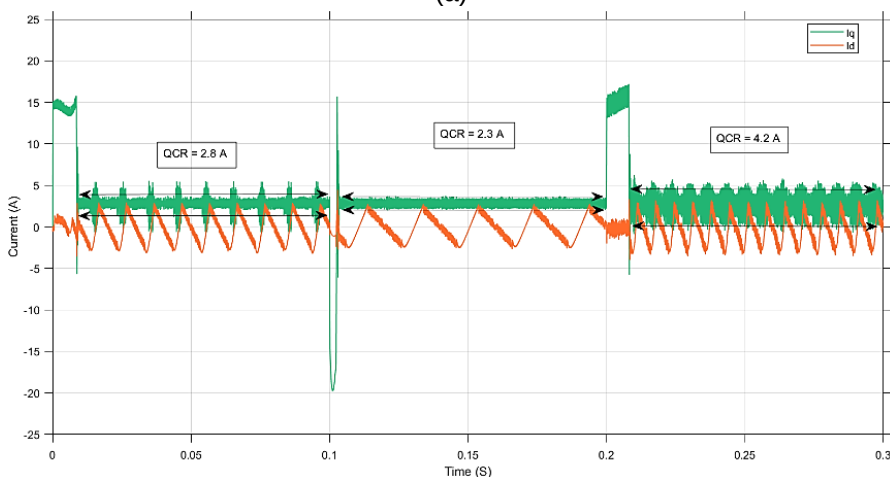


Source: Authors

Figure 10 – Comparison results with torque load of the proposed method and the traditional method for current ripples (a) Traditional method (b) Proposed method.



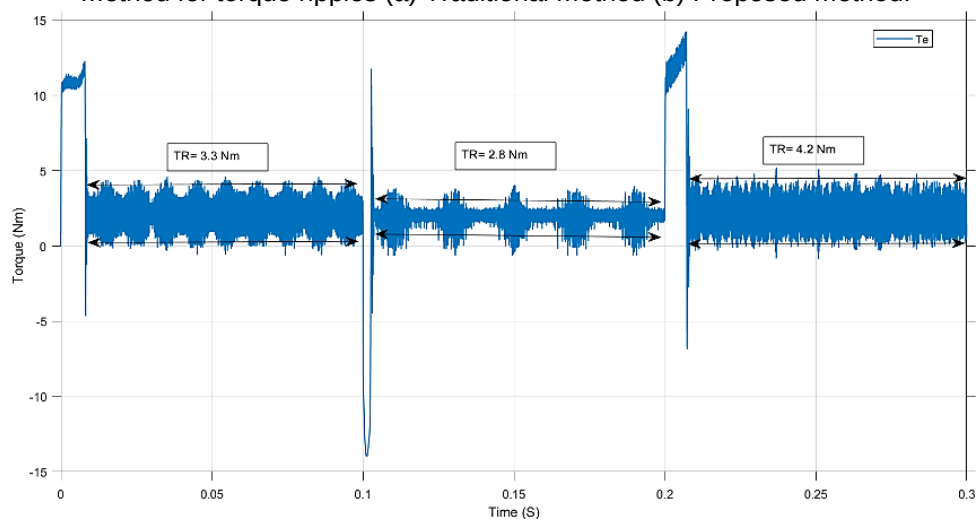
(a)



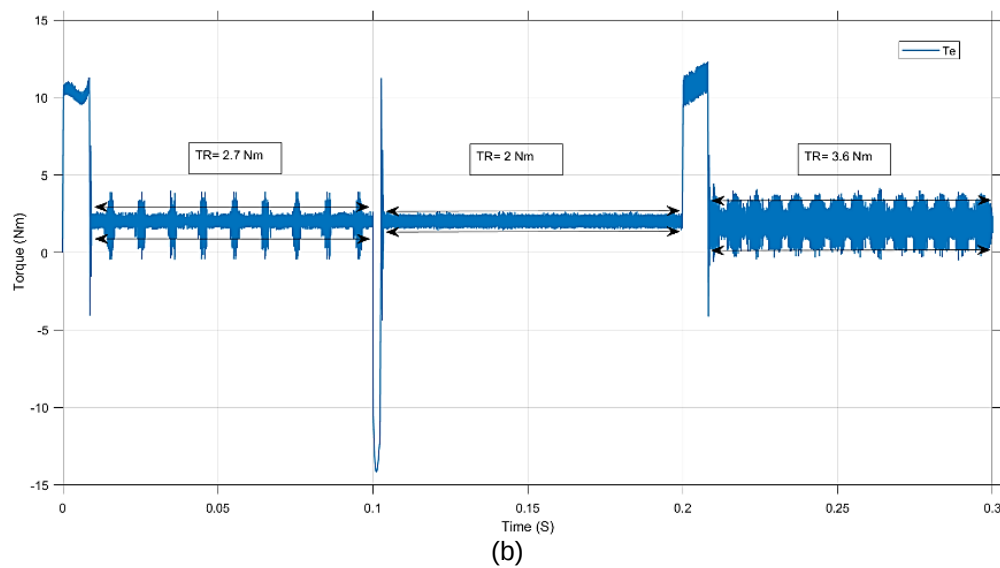
(b)

Source: Authors

Figure 11 – Comparison results with torque load of the proposed method and the traditional method for torque ripples (a) Traditional method (b) Proposed method.



(a)



Source: Authors

The Simulink model designed is introduced in this research to explore and evaluate the innovative approach to PMSM control outlined in this study. This Simulink model shown in Figure 5, the application of the Research Method section (3rd section). The “3-phase inverter” bloc, “abc2dq” bloc and the “PMSM” bloc are defined in subsection 3.1. The “MPC” bloc includes the details of subsections 3.2 to 3.6, the “SM-Flux weakening” bloc is designed as a subsystem that contains the details of subsection 3.7, the “Speed Control SM” bloc is sliding mode exponential for speed control which is considered in subsection 3.8.

In Figure 6, the dynamic speed tracking performance resulting from both the MPC and SMC is depicted. The speed profile comprises an initial setpoint of 1000 RPM from 0 to 0.1 seconds, a subsequent reduction to 500 RPM from 0.1 to 0.2 seconds, and a subsequent increase to 1500 RPM from 0.2 seconds onwards also in Figure 9, with the same trajectory by applying an external torque $T_l = 2$ Nm. The results demonstrate the capability of both the MPC and SMC controllers to accurately track the desired speed, also the graph indicate that the implementation of MPC leads to a reduction of the chattering in the common first order SM speed controllers.

In Figure 7, a comparison between the proposed method and a traditional approach is presented in terms of current ripples. The proposed method showcases a remarkable reduction in current ripple (QCR), with reductions of 11.76%, 8%, and 4.65% in the first, second, and third regions, respectively, when compared to the traditional method. This significant improvement is attributed to



the incorporation of the SM-Flux Observer in the proposed method, which effectively mitigates current ripples.

Figure 8, illustrates the electromagnetic torque ripple (TR) for both the proposed and traditional method. This reduction in torque ripple is achieved through the use of the proposed method, resulting in reductions of 17.39%, 10.52%, and 9.67% in the first, second, and third regions, respectively. Overall, this substantial improvement in torque ripples reduction is primarily attributed to the integration of the SM-Flux Observer within the proposed method, effectively mitigating the adverse effects of torque ripples.

In Figure 10, a comparison is illustrated between the suggested technique and a conventional approach regarding current ripples with the external torque load. The proposed method demonstrates a noteworthy decrease in current ripple (QCR), achieving reductions of 33.33%, 28.12%, and 14.28% in the first, second, and third regions, respectively, when contrasted with the conventional method. This notable enhancement is credited to the integration of the SM-Flux Observer in the proposed approach, effectively addressing and reducing current ripples.

Overall, in Figure 11, the electromagnetic torque ripple (TR) is depicted for both the proposed and conventional methods. The proposed approach successfully minimizes torque ripple, leading to reductions of 18.18%, 28.57%, and 14.28% in the first, second, and third regions, respectively. This considerable enhancement in torque ripple reduction is predominantly ascribed to the incorporation of the SM-Flux Observer in the proposed method, effectively alleviating the undesirable impact of torque ripples.

These results underscore the effectiveness of the proposed method, which combines (MPC), Sliding Mode Control (SMC), and Sliding Mode Observer (SMO) to accurately track the desired speed even when applying an external torque load and considerably mitigate current and torque ripples.

5 CONCLUSION

The simulation results have shown that the combination of SMC and MPC can be used to improve the performance of a surface-mounted PMSM. The results show that MPC can reduce the chattering of the SMC speed regulator and the SM



flux weakening observer can reduce the QCR and TR significantly. This is a major benefit as it allows for a smoother and more efficient operation of the system.

In conclusion, this study underscores the unique advantages of integrating SMO and MPC, showcasing not only their merits but also the cooperative improvements achieved through their combination. The simulation results confirm that integrating SMO and MPC effectively reduces chattering in speed regulation and minimizes current and torque ripples, contributing to smoother and more efficient PMSM operation. This contribution is an advancement in the control of PMSMs and could have significant implications for their use in a variety of industrial, commercial applications and especially for electric vehicles considering internal disturbances such the parameter variations, the sudden surges of current, and external disturbances like torque variations.

LIMITATION AND FUTURE RESEARCH:

The combination of SMC and MPC for PMSM control has the potential to provide superior performance compared to traditional control techniques. By combining the robustness of SMC with the predictive capabilities of MPC, it is possible to achieve better control of PMSM motors in a wide range of operating conditions. This could lead to improved efficiency, reduced vibration, and improved torque response. In the future, research into the use of SMC and MPC for PMSM control will focus on developing more efficient and robust control algorithms. This will involve exploring different combinations of SMC and MPC techniques, as well as investigating the use of advanced optimization techniques such as genetic algorithms and particle swarm optimization. Additionally, research will focus on developing methods for tuning the parameters of the control algorithms to achieve the best performance as a many different techniques applied by estimation strategies in the failure of the sensors (Kethiri; Charrouf, 2023a). The use of SMC and MPC for PMSM control is an exciting area of research that has the potential to revolutionize the way PMSM motors are controlled. With further research and development, the combination of SMC and MPC will likely become the standard for controlling PMSM motors in the future.



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